"CALCULUS"

FUNCTION

CHAPTER → 01

Function: A function is a rule that associates a unique output with each input. If the input is denoted by x, then the output is denoted by $f(x)$.

This output is sometimes called the value of f at x or the image of x under f. Sometimes we denote the output by a single letter y and write–

$$y = f(x).$$

Independent and Dependent Variables:

The variable x is called the independent variable (or argument) of f.

The variable y is called the dependent variabe of f.

Real valued Function :

The functions in which the independent and dependent variables are real numbers are called real valued function of a real variable.

e.g. — $f(x) = 4x$

$f(1) = 4$

The vertical Line Test :

A curve in the xy-plane is the graph of some function f, if and only if no vertical line intersects the curve more than once.

Exmp: The graph of the equation

$$x^2+y^2=25$$

$$\Rightarrow y = \pm\sqrt{25-x^2}$$

This equation does not define y as a function of x, because the right hand side is "multiple valued" in the sense that values of x in the interval $(-5,5)$ produce two corresponding values of y. However, we can regard the equation as the union of two equation —

$$y=\sqrt{25-x^2} \quad \text{and} \quad y=-\sqrt{25-x^2}$$

Domain and Range:

If $y = f(x)$, then the set of all possible inputs (x- values) is called the domain of f and the set of outputs (y-values) that results when x-varies over the domain is called the range of f.

Exmp:

i) $y = x^2$

The domain of the given function is $(-\infty, \infty)$ or

or $-\infty < x < \infty$

ii) $y = x^2, x \geq 2$

The domain of the given function is $[2, \infty)$

or $2 \leq x < \infty$

iii) $y = x^2$

$\Rightarrow x = \sqrt{y}$

As x varies over the domain of f the values of range of f vary over the interval, $0 \leq y < \infty$ or $[0, \infty)$.

iv) $y = x^2, \quad x \geq 2$

$\Rightarrow x = \sqrt{y}$

As x varies over the domain of the function $f(x) = x^2$, $x \geq 2$, the values of y vary over the interval,

$4 \leq y < \infty$ or $[4, \infty)$.

Natural domain:

If a real valued function of a real variable defined by a formula and if no domain is stated explicitly, then it is to be understood that the domain consists of all real number for which the formula yields a real value. This is called the natural domain of the function.

Exmp:

* Find the natural domain and range of the given following function —

(a) $f(x) = \frac{x+1}{x-1}$ (b) $f(x) = \frac{x^2-4}{x-2}$ (c) $f(x) = \frac{x^2}{x^2-1}$

Solve:

a) The domain of f is $(-\infty, 1) \cup (1, \infty)$ or $R - \{1\}$

Now, Put, $y = \frac{x+1}{x-1}$

$\Rightarrow x+1 = xy - y$

$\Rightarrow xy - x = y + 1$

$\Rightarrow x = \frac{y+1}{y-1}$

It is evident from the right side of this equation that $y = 1$ is not in the range. Otherwise, we would have a division by zero (0).

So, the range of the function is —

$(-\infty, 1) \cup (1, \infty)$ or $R - \{1\}$

b) The given function is defined for all x,
~~exp.~~ except $x \neq 2$,

So, the domain of f is $(-\infty, 2) \cup (2, \infty) \Rightarrow R - \{2\}$

Now, put, $y = \frac{x^2-4}{x-2}$ $\boxed{x = y-2}$

$\Rightarrow y = x+2$, provided that $x \neq 2$

$\therefore$ The function is not defined for $y = 4$, since $x \neq 2$

Hence, the range of the function is

$(-\infty, 4) \cup (4, \infty) \Rightarrow R - \{4\}$.

c) The given function is $f(x) = \frac{x^2}{x^2-1}$

The function is defined for all x, except $x = \pm 1$

$\therefore$ The domain is $R - \{1, -1\}$.

Now, Put, $y = \frac{x^2}{x^2-1}$

$\Rightarrow x^2 y - y = x^2$

$\Rightarrow x^2 y - x^2 = y$

$\Rightarrow x^2 = \frac{y}{y-1}$

$\Rightarrow x = \pm \sqrt{\frac{y}{y-1}}$

That the rightside of the above equation is not defined in the interval $[0, 1]$

So, the range of the given function is $R - [0, 1]$

Inverse Function:

If the function f and g satisfy the two conditions $g(f(x)) = x$ for every x in the domain of f. $f(g(y)) = y$ for every y in the domain of g. Then we say that f and g are inverse function.

Moreover, if f is an inverse function of g, then g is an inverse function of f.

Normally we denote the inverse function by $g = f^{-1}$ (for the case of g and f)

Exmp: Show that inverse of $f(x) = 2x$ is $f^{-1}(x) = x/2$

Solve: Here, $f(f^{-1}(x)) = f(x/2) = 2 \cdot \frac{x}{2} = x$

$$f^{-1}(f(x)) = f^{-1}(2x) = \frac{2x}{2} = x$$

So, the function are inverses.

Denoting Inverse Function:

If a function f has an inverse, then the inverse is unique. The inverse of a function f is commonly denoted by f^{-1}.

Remember that, f^{-1} should be interpreted as the inverse of f and never as the reciprocal $\frac{1}{f}$. So $f^{-1} \neq \frac{1}{f}$.

Algebraic Function:

Function that can be constructed from polynomials applying finitely many algebraic operations (addition, subtraction, division, root extraction etc.) are called algebraic function.

e.g: $f(x) = \sqrt{x^2-4}$; $f(x) = x^{2/3}.(x+2)^2$.

Trigonometric Function:

The sine, cosine, tangent, cotangent, secent, cosecent of a possitive acute angle θ can be difined as ratios of the sides of right angle. We call sin, cos, tan, cot, sec, cosec the trigonometric functions.

Examp: $\sin\theta = 1/2$

Exponential Function:

A function of the form $f(x) = b^x$, where $b>0$ and $b \neq 1$, is called an exponential function with base b.

Examp: $f(x) = 2^x$.

Exponential function are continuous and have one of the basic two shapes shown in ~~fuge~~ figure

Sin hx ~~fn~~ has a domain $(-\infty,\infty)$ and range of $(-\infty,\infty)$ where as $\cosh x$ has a domain of $(-\infty,\infty)$ and range of $[1,\infty)$.

e.g.: Find the domain and range of the following functions and confirm your result graphically —

(a) $f(x) = \sqrt{1-x}$ (b) $f(x) = \begin{cases} x^2; & \text{if } x < 0 \\ 1+x; & \text{if } 0 \le x \le 1 \end{cases}$

(c) $f(x) = -x^2$

Solution – (a)

The given function is $f(x) = \sqrt{1-x}$, Here $\sqrt{1-x}$ is difined only when $1-x \geq 0$; that is only when $x \leq 1$

So, the domain is $(-\infty, 1]$

Now, consider $y = \sqrt{1-x}$ since $\sqrt{1-x}$ can not be negative, therefore the value of y must be nonnegative.

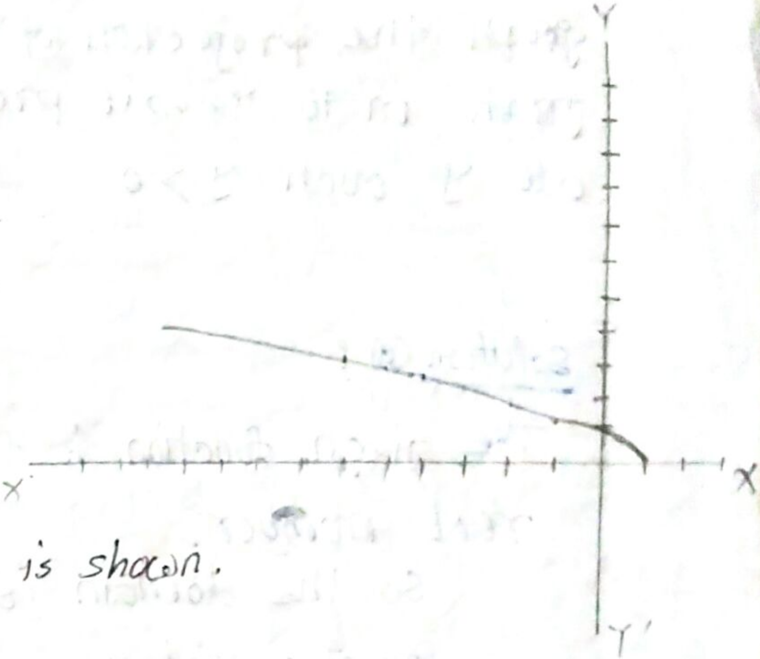
Hence, the range consists of all non-negative numbers.

So the range is $[0,\infty)$.

Now, we have $y = \sqrt{1-x}$

Contruct a table with the coresponding values of x and y —

x	1	-1	-2	-3	-4	-5	-6	0
y	0	$\sqrt{2}$	$\sqrt{3}$	2	$\sqrt{5}$	$\sqrt{6}$	$\sqrt{7}$	1



The graph of the function is shown.

$y = \sqrt{1-x}$

$\Rightarrow x = 1 - y^2$

The graph of the upper half is parabolla.

$x = 1 - y^2$

which is consists with our result.

Solution(b):

Here, $f(x)$ is determined by two different formulas. The first formula applies when x is negative, that is $x < 0$ and the second when $0 \leq x \leq 1$.

Hence, the domain is given by —

$(-\infty, 0) \cup [0, 1]$

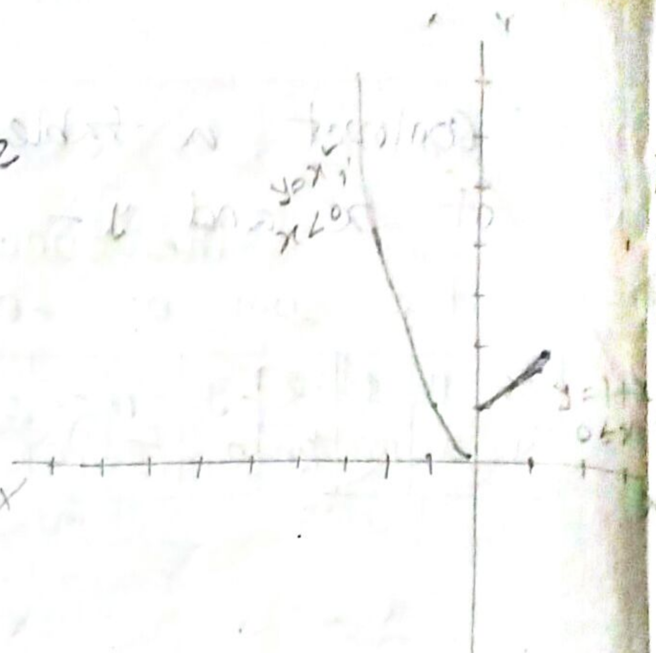
$= (-\infty, 1]$

And Range is given by —

$(0, \infty) \cup [1, 2]$

$= (0, \infty)$.

The function is exhibited by the graph the projection of the graph on to Y-axis produces all y such $y > 0$.



Solution (c) :

The given function is $f(x) = -x^2$, Here, $-x^2$ is defined by all real number.

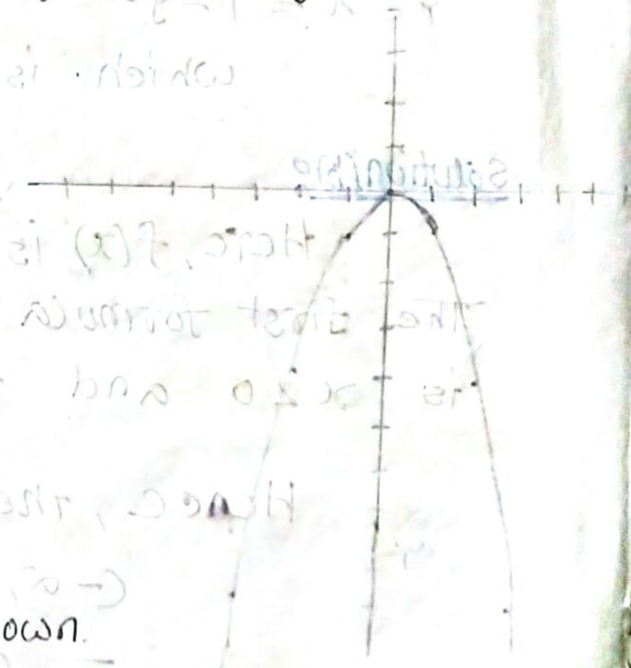
So, the domain is R or $(-\infty, \infty)$

Now, consider $y = -x^2$ since $-x^2$ always be negative with zero. Therefore, the value of y must be negative with zero.

Hence, the range consists off all negative numbers with zero.

So the range is $(-\infty, 0]$

Now we have $y = -x^2$



controvert a table with the corresponding value of x and y-

x	0	1	-1	2	-2	3	-3
y	0	-1	-1	-4	-4	-9	-9

The graph of the function is shown.

$y = -x^2 \Rightarrow x^2 = -y$

$\Rightarrow x = \sqrt{-y}$

The graph of the lower half is the parabolla $x = \sqrt{-y}$ which is consistent with our result.

2.8. Classification of Functions

(I) Even and Odd Functions

Let $f(x)$ be a function defined in a domain $D(\in R)$ where D is such that $x \in D \Rightarrow -x \in D$. The function $f(x)$ is said to be an *even function* if $f(-x) = f(x)$, for all $x \in D$ and $f(x)$ is called to be an *odd function* if $f(-x) = -f(x)$ for all $x \in D$.

The graph of an even function is symmetrical about the axis of y while the graph of an odd function is symmetrical in opposite quadrants.

Every function can be expressed as the sum of an even an odd funtion.

It should be noted that *inverse* of an even function is not defined.

2.11 Miscellaneous Worked out Examples.

Ex. 1 Show that $f(x) = \log\left(x + \sqrt{1 + x^2}\right)$ is an odd function of x.

Solution : $f(x) + f(-x) = \log\left(x + \sqrt{1 + x^2}\right) + \log\left(-x + \sqrt{1 + x^2}\right)$

$$= \log\left\{\left(x + \sqrt{1 + x^2}\right)\left(-x + \sqrt{1 + x^2}\right)\right\}$$

$$= \log\left\{1 + x^2 - x^2\right\} = \log 1 = 0$$

i.e., $f(x) = -f(-x)$

Thus $f(x)$ is an odd function of x.

Ex. 2. Prove that any function of x, defined for all real values of x, can be expressed as the sum of an even and an odd function of x.

Solution : Let, $f(x)$ be any function of x defined for all real values of x.

We can write, $f(x)=\frac{1}{2}\{f(x)+f(-x)\}+\frac{1}{2}\{f(x)-f(-x)\}$

$$=\phi(x)+\psi(x)\text{, say}$$

Now, as $\phi(x)=\frac{1}{2}\{f(x)+f(-x)\}$,

$$\phi(-x)=\frac{1}{2}\{f(-x)+f(x)\}=\phi(x),$$

So, $\phi(x)$ is an even function of x.

Also, $\psi(x)=\frac{1}{2}\{f(x)-f(-x)\}$

$$\psi(-x)=\frac{1}{2}\{f(-x)-f(x)\}=-\frac{1}{2}\{f(x)-f(-x)\}=-\psi(x),$$

so that $\psi(x)$ is an odd function of x.

Thus any function $f(x)$ of the real variable x can be expressed as the sum of an even function and an odd function of x.

2.11 Miscellaneous Worked out Examples.

Ex 1 Show that $f(x) = \log\left(x + \sqrt{1+x^2}\right)$ is an odd function of x.

Solution : $f(x) + f(-x) = \log\left(x + \sqrt{1+x^2}\right) + \log\left(-x + \sqrt{1+x^2}\right)$

$$= \log\left\{\left(x + \sqrt{1+x^2}\right)\left(-x + \sqrt{1+x^2}\right)\right\}$$

$$= \log\left\{1 + x^2 - x^2\right\} = \log 1 = 0$$

i.e., $f(x) = -f(-x)$

Thus $f(x)$ is an odd function of x.